PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Business Optimizer

Version: 1.0

1. Project Overview

- **Problem Statement:**
  In a saturated marketplace where consumer attention spans are declining quickly, businesses might struggle to establish a trustworthy brand that meet evolving customers' needs consistently. Traditional branding often fails to capture attention of customer immediately, leading to a gap between a company's core message and its target audience's perception (Indeed Editorial Team, 2025).
- **Target Domain:**
  Enabling rapid-fast UI/UX for prototypes and early-stage solution-based product development. The focus is on reducing the "Idea-to-Market" cycle for business by generating business strategy through AI-driven insights.
- **Proposed Solution Summary:**
  An AI-driven strategic engine that transforms raw market data and internal data into executable business strategy. By automating the transition from market analysis to designing marketing mix strategy, it successfully pivots from reactive data-gathering to proactive market leadership. This will replace manual research and human analysis with context-aware recommendations based on thorough and detailed analysis. Additionally, it can be one step above, predicting the potential challenges the business may face based on the context. This allows businesses to be prepared for sudden market changes, allowing them to follow the market trend comfortably. To make sure the strategy fits the user's business, users can upload their own data into the system before it generates marketing mix. Lastly, it can compare two different marketing strategies and provide its recommendations, making it a decision maker.

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2. Background & Business Objective

- **Background of the Problem:**
  Successful businesses must prioritize meeting customer needs through effective branding and following trends. As data is more easily accessible and attention spans of customer shorten, companies must stay present to get customer's trust. However, the sheer volume of unstructured data might be too time consuming to interpret, causing for slow market analysis. This can lead to the business not following the market trend, impacting profit. Additionally, a bad marketing mix strategy might lead to a business being seen as inconsistent, damaging trust between customers and business. As market becomes increasingly saturated, businesses also struggle to come up with new a creative strategy, to gain customers' attention and overshadow competitors (Indeed Editorial Team, 2025).
- **Importance of Solving This Issue:**
  To solve the gap between gathering data and strategic execution for business. In a saturated market, the ability to interpret market data and provide a strategic plan based on the context becomes a necessity. The business can save time by reducing the time it takes to come up with a suitable strategy, allowing them to follow the market trend. Business owners must have a high-level system that could analyze, make decisions and optimize business strategy to level the playing field.
- **Strategic Fit / Impact:**
  It aligns with the National Fourth Industrial Revolution (4IR) Policy, Strategy 13. Strategy 13 is about facilitating the adoption of 4IR technologies among local businesses. Business Optimizer is a tool specifically designed to help business adopt Artificial Intelligence (AI) without needing a massive budget. This lowers the barrier of entry for business to integrate AI into their daily operations (Kementerian Ekonomi, n.d.). It also aligns with Malaysia Digital (MD) by encouraging digital adoption among aspiring young entrepreneurs, companies and citizens (*Malaysia Digital Economy Corporation, n.d.)*.

3. Product Purpose

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3.1. Main Goal of the System:

To provide an AI-enabled analyst and decision maker. It aims to come up with the best business strategy based on the given market data. This is so that the business can attract customer attention, increasing profit.

3.2. Intended Users (Target Audience):

- Business Owners/ Entrepreneur
- Business Stakeholder
- Market Analyst

4. System Functionalities

4.1. Description:

The system operates as a high-speed generative engine for market analysis and business strategy planning. Using data given by the user, the system will be able to use GLM to interpret both unstructured and structured data. After performing market analysis, the system will suggest a marketing mix strategy with detailed reasoning behind every choice. It shows strong decision making by being able to compare two strategies.

4.2. Key Functionalities:

- **Market Analysis**

  Interpreted structured and instructed data provided by the user to understand the market change and potential opportunities and explain with context.
- **Marketing Mix Proposal**

  After market analysis, it will provide a contextual marketing mix strategy for the business to use. Each decision is also explained based on the context to ensure the plan is not outdated.
- **Vertex AI Search**

  Get context information based on region and industry. This is to provide context for market mix comparison.
- **Marketing Mix Comparison**

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Compare two different marketing strategies and pick the best one with reasoning. Further improvement suggestion will also be made.

- **Save Data**

  Able to save market data into database so that market analysis will have more data to work with

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

The LLM used is Z.ai GLM. The model used is ilmu-glm-5.1. This model is chosen because it provides a superior balance between reasoning capability and latency. In a market-facing application where users require fasr comparisons of marketing strategies, the 5.1 architecture offers faster time to first token compared to larger models. It does still while maintaining the logical depth required for business analysis.

4.3.2. **Prompting Strategy:**

The system employs a Zero shot prompting with Structural Constraints. It enforces structured synthesis prompting strategy. By providing a strict JSON schema, it enforce a multi-step reasoning chain where the model must first do a market analysis before giving a marketing mix proposal. This ensures all strategic recommendations are grounded in data-driven insights and not generic advice.

4.3.3. **Context & Input Handling**

Our system sets a limit of 10,000 character per save data. This ensures that the model doesn't lose track of the earlier parts of the data when processing.

4.3.4. **Fallback & Failure Behavior:**

Z.ai is required to provide a strict JSON schema, where it will undergo a parser check. Fast API backend will try to parse it into a Python object. If the JSON is deformed, the system will show an error message instead of crashing. From there, user can then try again by pressing the try again button. A sanity check will also be done on a quantitative result such as estimate revenue growth. If he number is too high, it will flag down and users will be asked to retry

5. User Stories & Use Cases

- **User Stories:** *As a business owner, I want a business strategy so that I can follow the latest market trend.*
- **Use Cases (Main Interactions):**

  Upload data : User paste their structured or unstructured data into firebase.

  Market Analysis: User then start analysis where the uploaded data is used in Z.ai to analyze and find business opportunities

  Marketing Mix Generation: Once Market Analysis in completed, it will be used to create a suitable Marketing Mix Strategy along with context-aware reasoning. The strategy will then be saved into database
- **Use Case Diagram:**

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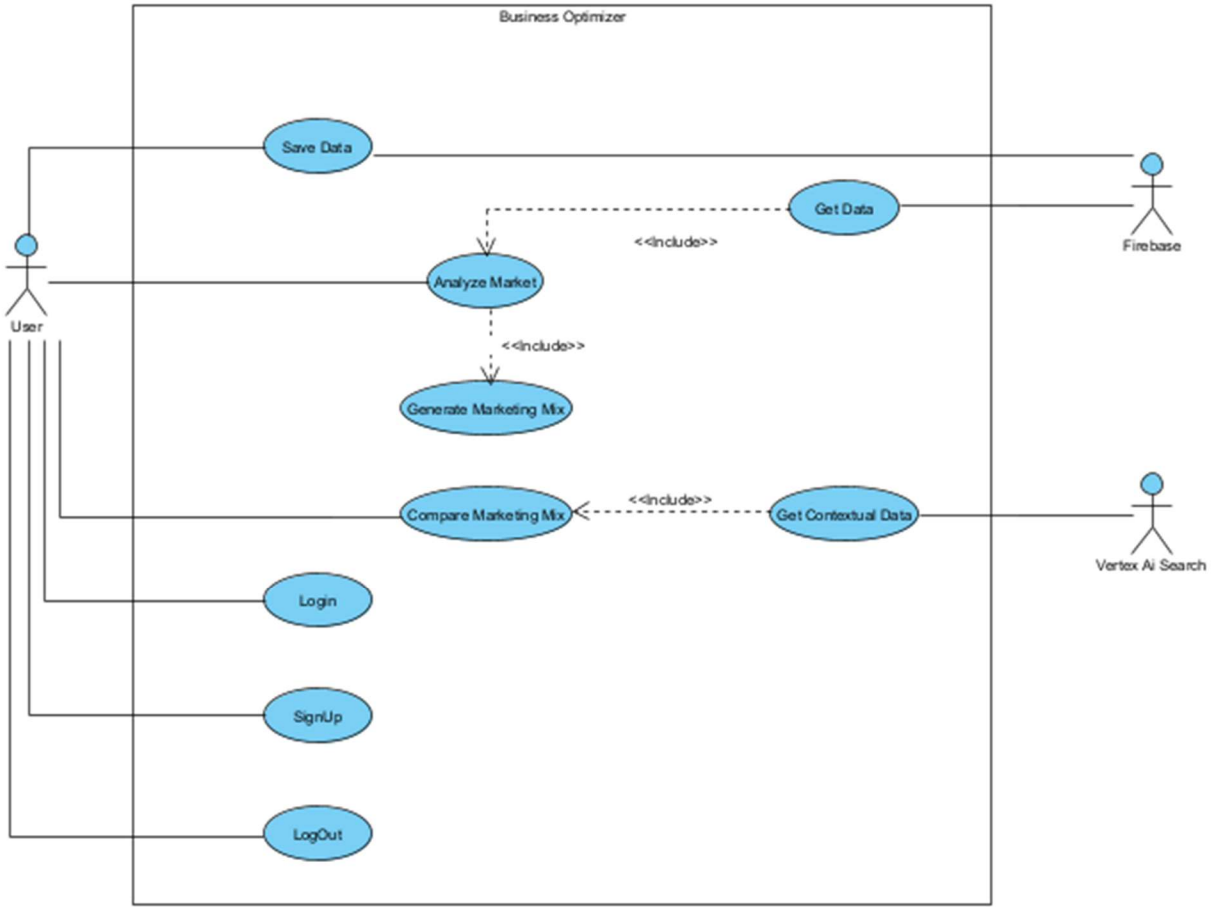

6. Features Included (Scope Definition)

- Marketing Mix Generation, Analysis and Comparison using Z.ai
- A navigable UI frontend
- Save data and get data feature
- Login, Logout, Sign Up
- Vertex AI Search for context information

7. Features Not Included (Scope Control)

- Implementation of user generation history, where users can access all their previous generations

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- An emergency rollback feature and golden version
- Automatically saving comparison, generation and analysis into firebase
- Fallback to older model once current Z.ai model limit has been reached
- Profile Management

8. Assumptions & Constraints

- **LLM Cost Constraint:**
  We estimate an average session to utilize approximately 4,000 tokens. We implemented Strict Output Schema Enforcement so that it returns JSON object and prevent conversational response. This minimizes output token generation.
- **Technical Constraints:** Our system is limited to the Android operating system. Users on iOS or Desktop environments are currently out of scope as it uses Android SDK and Kotlin.
- **Data Availability:** For generating marketing mix, the system currently only rely on uploaded data. If the data uploaded is limited, the market analysis and marketing mix generation will also be limited.

9. Risks & Questions Throughout Development

- **Data Used:** Would outdated data that was uploaded long ago affect the marketing mix generation?
- **Context-Aware Reasoning:** Is there enough context for the AI to create an accurate context-aware reasoning?
- **AI hallucination**: Would the AI hallucinate and give accurate responses?

10. References

Indeed Editorial Team. (2025, December 17). *6 Challenges Most Businesses Face*. Indeed Career Guide. https://www.indeed.com/career-advice/career-development/challenges-businesses-face

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